

CS3 Case Study Rubric: Sentiment Analysis of Restaurant Reviews

Ciara Fasullo

DS 4002

06 May 2026

Formatting	• Clear organization with labeled sections • Concise and readable writing • Code runs without errors • All required components are included • Includes clear explanation of results in written form
Task Overview	Goal: • Build a model that classifies restaurant reviews as positive, neutral, or negative using only the text of the reviews • Evaluate model performance and explain key findings in a short written summary
Data Preparation	Goal: Prepare the dataset for modeling • Remove rows with missing or empty review text • Ensure rating values are valid and numeric • Convert ratings into sentiment categories: ◦ Below 3 → Negative ◦ Equal to 3 → Neutral ◦ Above 3 → Positive • Clean text by: ◦ Lowercasing ◦ Removing punctuation/non-alphabetic characters ◦ Removing common stop words Meets Expectations: • Data is cleaned and usable • Sentiment labels are correctly assigned • Text preprocessing is clearly applied
Feature Creation	Goal: Convert text into numerical features • Use TF-IDF to represent review text • Apply consistent feature settings across the dataset ◦ (e.g., same vocabulary and parameters for all data) Meets Expectations: • TF-IDF is implemented correctly • Features are appropriate for model input
Modeling	Goal: Train a model to classify sentiment • Use a classification model (Logistic Regression or Naive Bayes) • Split data into training and testing sets • Address class imbalance (e.g., through class weighting) Meets Expectations: • Model is trained correctly • Training/testing split is applied

	• Model choice is appropriate
Evaluation	Goal: Assess model performance • Report accuracy on test data • Examine performance across sentiment classes (e.g., confusion matrix or class-level accuracy) • Identify strengths and weaknesses of the model Meets Expectations: • Accuracy is calculated correctly • Performance is interpreted (not just reported) • Limitations are acknowledged
Interpretation & Insights	Goal: Explain what the model reveals • Discuss patterns in predictions • Identify challenges (e.g., neutral sentiment, classification) • Explain impact of class imbalance • Reflect on why the model performs as it does Meets Expectations: • Insights go beyond surface-level • Clear explanation of results • Demonstrates understanding of model behavior
Success Criteria	• A strong model achieves ~70% accuracy or higher • The full pipeline (cleaning → features → model → evaluation) is completed • The student provides a clear explanation of results and limitations • The student demonstrates understanding of both model performance and its limitations